

Direct Oral Anticoagulant

Outline

- Introduction
- Indication & contraindication
- Dose
- Extreme body weight
- Renal & liver impairment
- Drug interactions
- Monitoring parameters
- Reversal of DOAC



Dabigatran



Apixaban

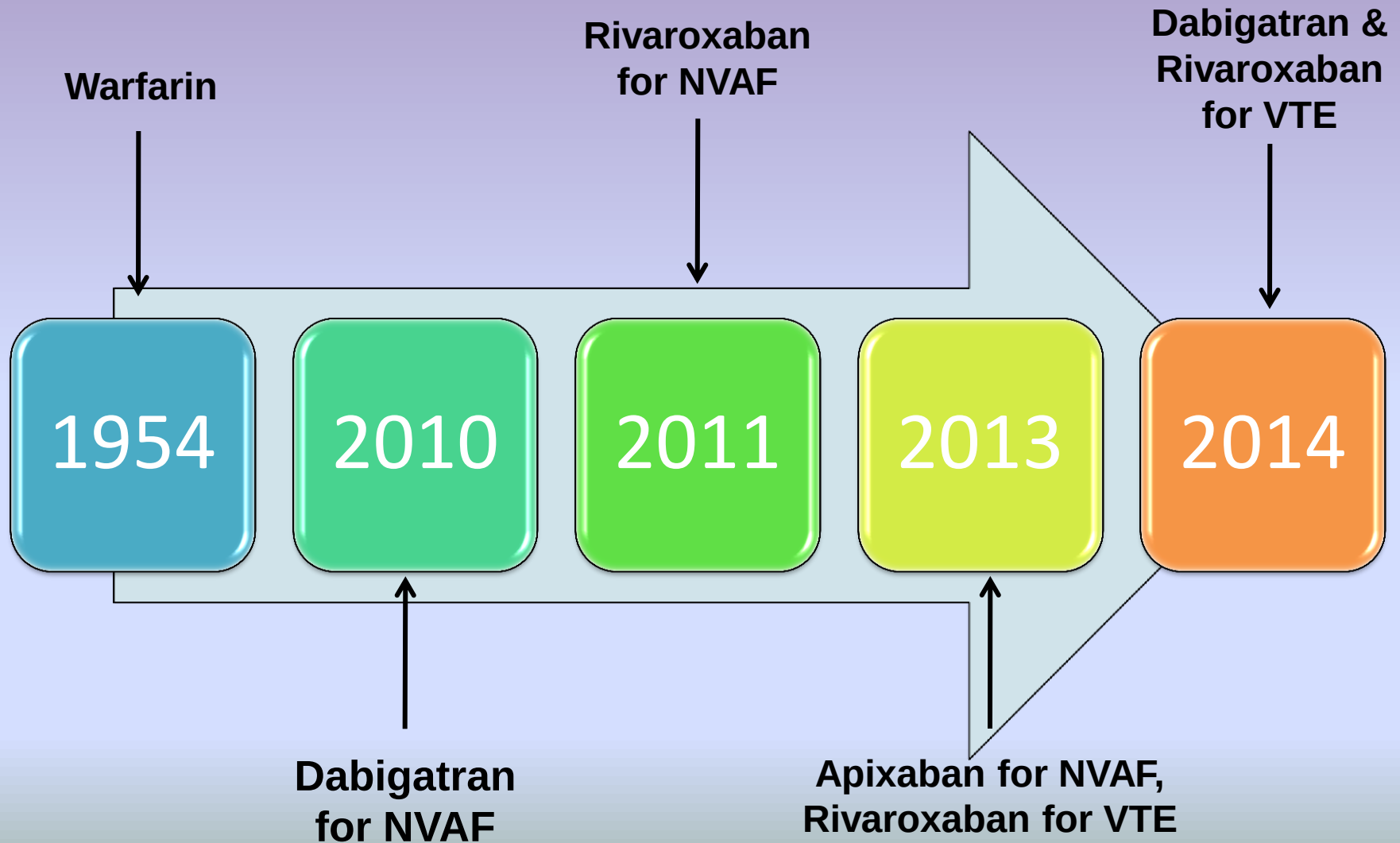


Rivaroxaban



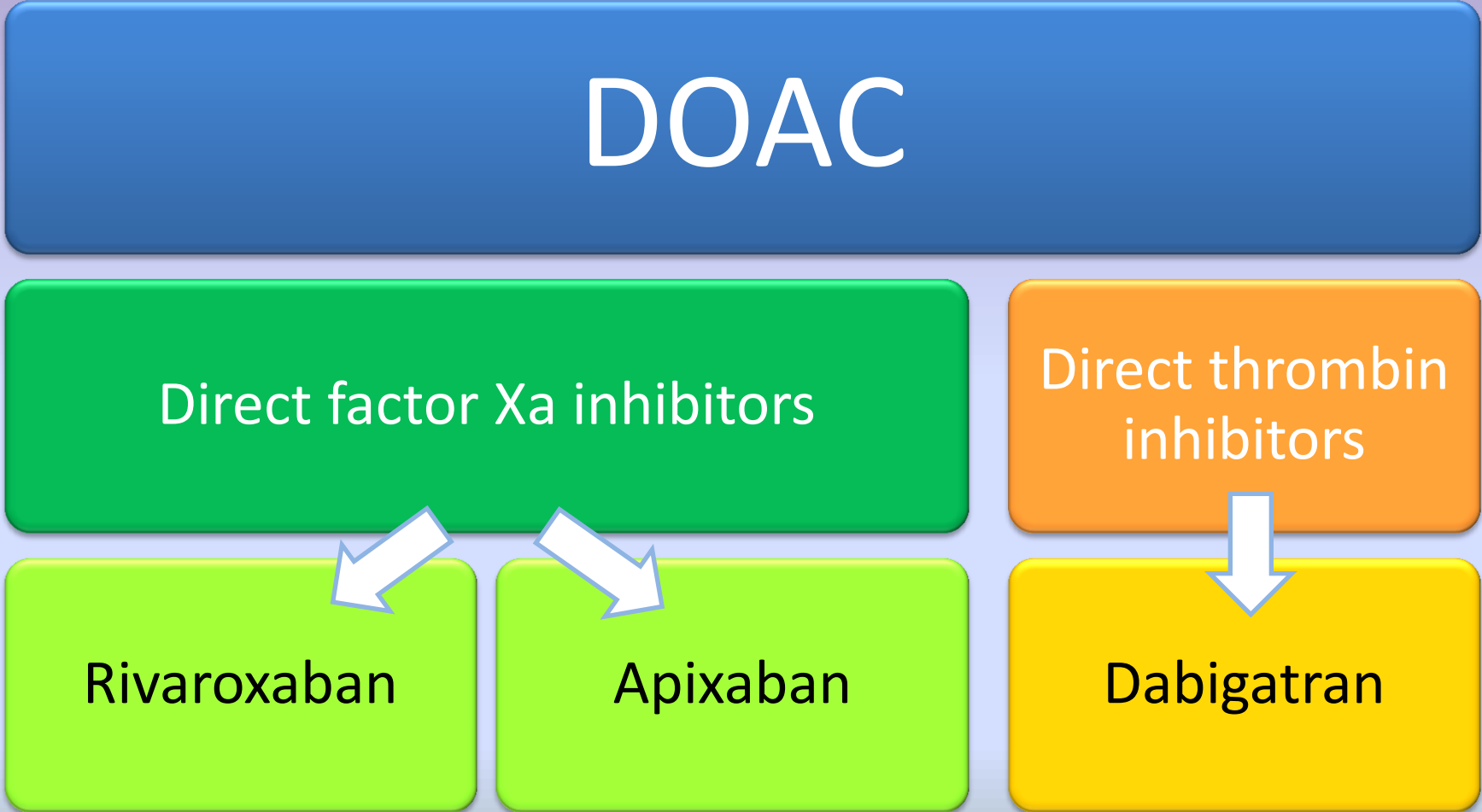
Edoxaban

Timeline of Oral Anticoagulant Development



DOAC Category

DOAC



```
graph TD; DOAC[DOAC] --> DXa[Direct factor Xa inhibitors]; DOAC --> DTI[Direct thrombin inhibitors]; DXa --> Rivaroxaban[Rivaroxaban]; DXa --> Apixaban[Apixaban]; DTI --> Dabigatran[Dabigatran];
```

The diagram is a hierarchical flowchart. At the top is a blue rounded rectangle labeled 'DOAC'. Below it are two rounded rectangles: a green one on the left labeled 'Direct factor Xa inhibitors' and an orange one on the right labeled 'Direct thrombin inhibitors'. From the green box, two white arrows point down to two light green rounded rectangles labeled 'Rivaroxaban' and 'Apixaban'. From the orange box, a single white arrow points down to a yellow rounded rectangle labeled 'Dabigatran'.

Direct factor Xa inhibitors

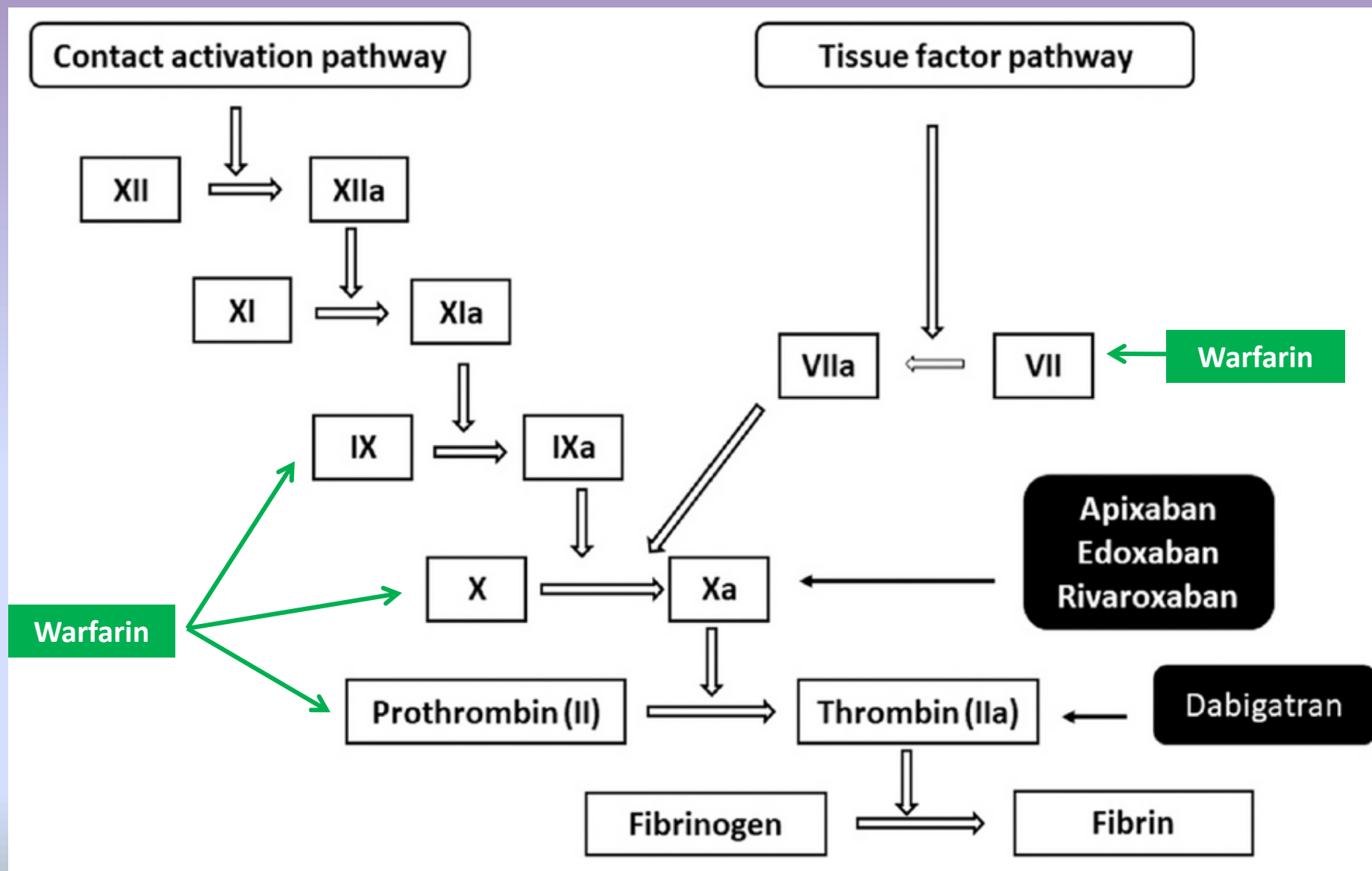
Rivaroxaban

Apixaban

Direct thrombin
inhibitors

Dabigatran

DOAC site of action in coagulation cascade



Pharmacokinetic Profile

Agent	Warfarin	Dabigatran	Rivaroxaban	Apixaban
Target	IIa, VIIa, IX, Xa	IIa	Xa	Xa
Peak effect	4-5 days	1.5-3 h	2-4 h	1-3 h
Half-life (h)	40	12-17	5-9	9-14
Renal element (%)	None	80	33	25
Dialyzability	No	Yes	No	No
Interactions	Many	P-gp	3A4, P-gp	3A4, P-gp
Monitoring	Yes	No	No	No
Antidote	Vitamin K	Idarucizumab	No	No
Laboratory monitoring	INR	aPTT, diluted TT	PT, anti-Xa	PT, anti-Xa

P-gp=Glycoprotein; 3A4=Cytochrome P4503A4; aPTT=Activated partial thromboplastin time; TT=Thrombin time; PT=Prothrombin time; INR=International normalised ratio

Nevine A. Kassim. The new, direct, target-specific oral anticoagulants. December 14, 2020

Indication

- Prevention of stroke in non-valvular atrial fibrillation
- Prevention of venous thromboembolism in total knee/ hip replacement surgery
- Treatment of Deep Vein Thrombosis (DVT) / PE
- Prevention of Recurrent DVT/ Pulmonary Embolism

Contraindication

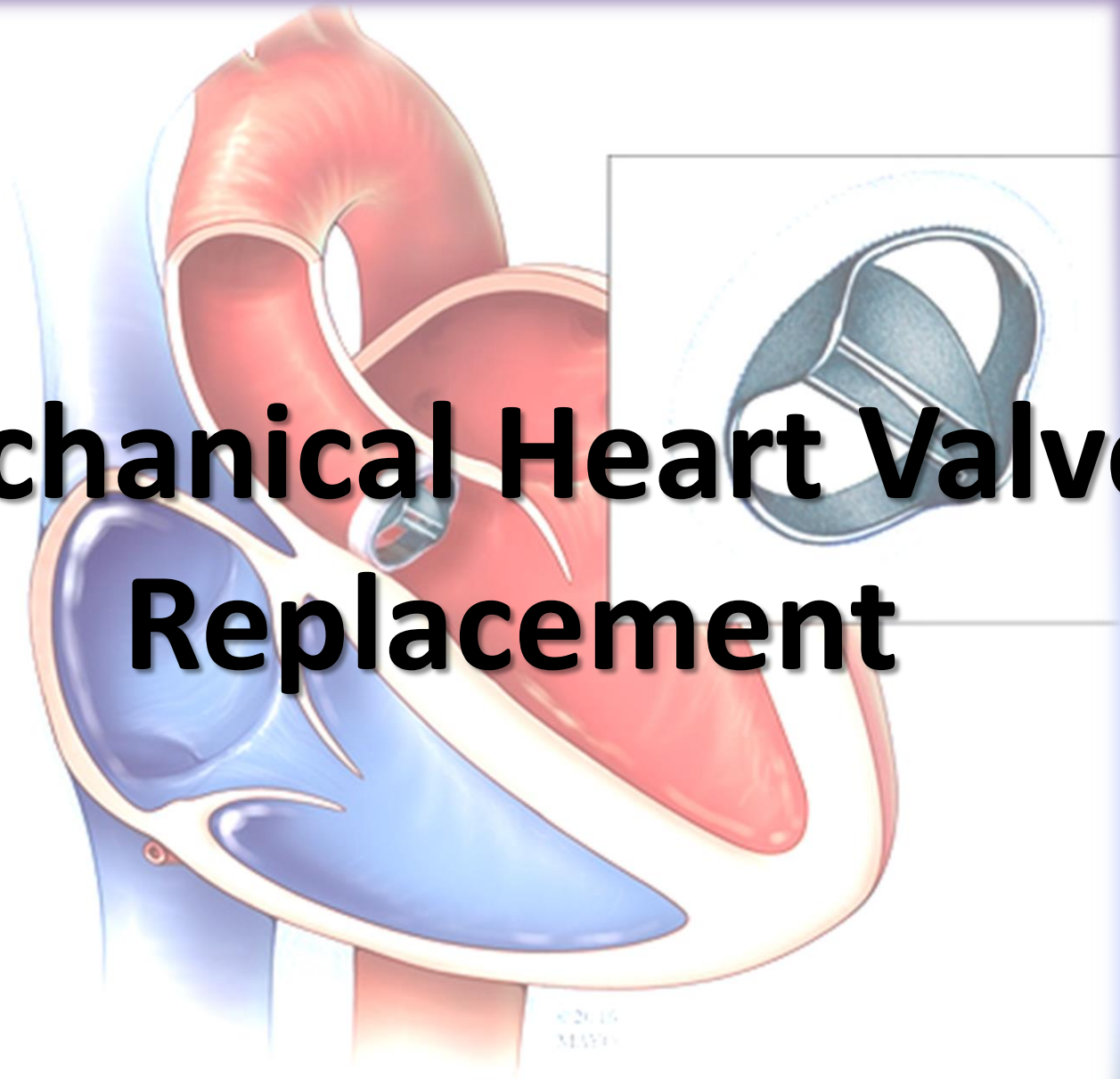
- ☐ Hypersensitivity to the active substance or to any of the excipients
- ☐ Severe liver/renal disease
- ☐ Mechanical heart valve
- ☐ Active bleeding (< 6 months) (e.g intracranial hemorrhage, recent GI bleed)
- ☐ Lesion or condition at significant risk of major bleeding (e.g GI ulceration, vascular aneurysm, arteriovenous malformations)
- ☐ Pregnancy or breastfeeding

Reference:

1.AC-MTAC Protocol 2nd Edition

2. Nevine A. Kassim. *The new, direct, target-specific oral anticoagulants*. December 14, 2020

Mechanical Heart Valve Replacement



RE-ALIGN trial (Dabigatran Versus Warfarin in Patients With Mechanical Heart Valves)

- DOACs is contraindicated in patients with mechanical heart valves.
- The study was terminated early, as patients with mechanical prosthetic heart valves experienced excess thromboembolic and bleeding events.
- VKA remains the preferred agent for this patient population.
[ACCP/ AHA guideline]





Pregnancy and Breastfeeding



Contents lists available at [ScienceDirect](http://www.sciencedirect.com)

Thrombosis Research

journal homepage: www.elsevier.com/locate/thromres

Short Review

Efficacy and safety of direct oral anticoagulants during pregnancy; a systematic literature review

Heleen Lameijer^{a,b,*}, Jan J.J. Aalberts^a, Dirk J. van Veldhuisen^a, Karina Meijer^c,
Petronella G. Pieper^a

^a Department of Cardiology, University Medical Centre Groningen, University of Groningen, Groningen, the Netherlands

^b Department of Emergency Medicine, Medical Centre Leeuwarden, Leeuwarden, the Netherlands

^c Department of Haematology, University Medical Centre Groningen, University of Groningen, Groningen, the Netherlands

236 cases of DOAC use during pregnancy were reported.
Rivaroxaban was the most reported DOAC (n=178).
DOACs were mostly used for prophylaxis or treatment of venous thromboembolism.

Findings

DOACs were discontinued within the first 2 months of in 84% of pregnancy

Pregnancy outcome data were available for 140 pregnancies.

39 pregnancies were electively terminated.

101 pregnancies - total miscarriage rate was 31% (n=31) and live birth rate was 68% (n=69, 1 missing).

Foetal and neonatal abnormalities were reported in 8 pregnancies.

Presence or absence of thrombotic and bleeding complications was reported in 18% of cases (n=42)

Conclusion

- The limited available evidence raises concern regarding embryo-foetal safety, because of a high incidence of miscarriages and a 4% rate of anomalies with the use of rivaroxaban.
- The current limited evidence justifies avoidance of DOACs in pregnant women.

Breastfeeding



- No clinical data on whether DOAC are excreted into breastmilk.
- International Society on Thrombosis and Haemostasis recommended that breastfeeding women should not be treated with DOACs in the absence of evidence for their safe use in this situation.

Reference:

-Lactmed

-Product insert for Dabigatran, Rivaroxaban, and Apixaban

-H. Cohen. Journal of Thrombosis and Hemostasis. Management of direct oral anticoagulants in women of childbearing potential: guidance from the SSC of the ISTH

Dosing

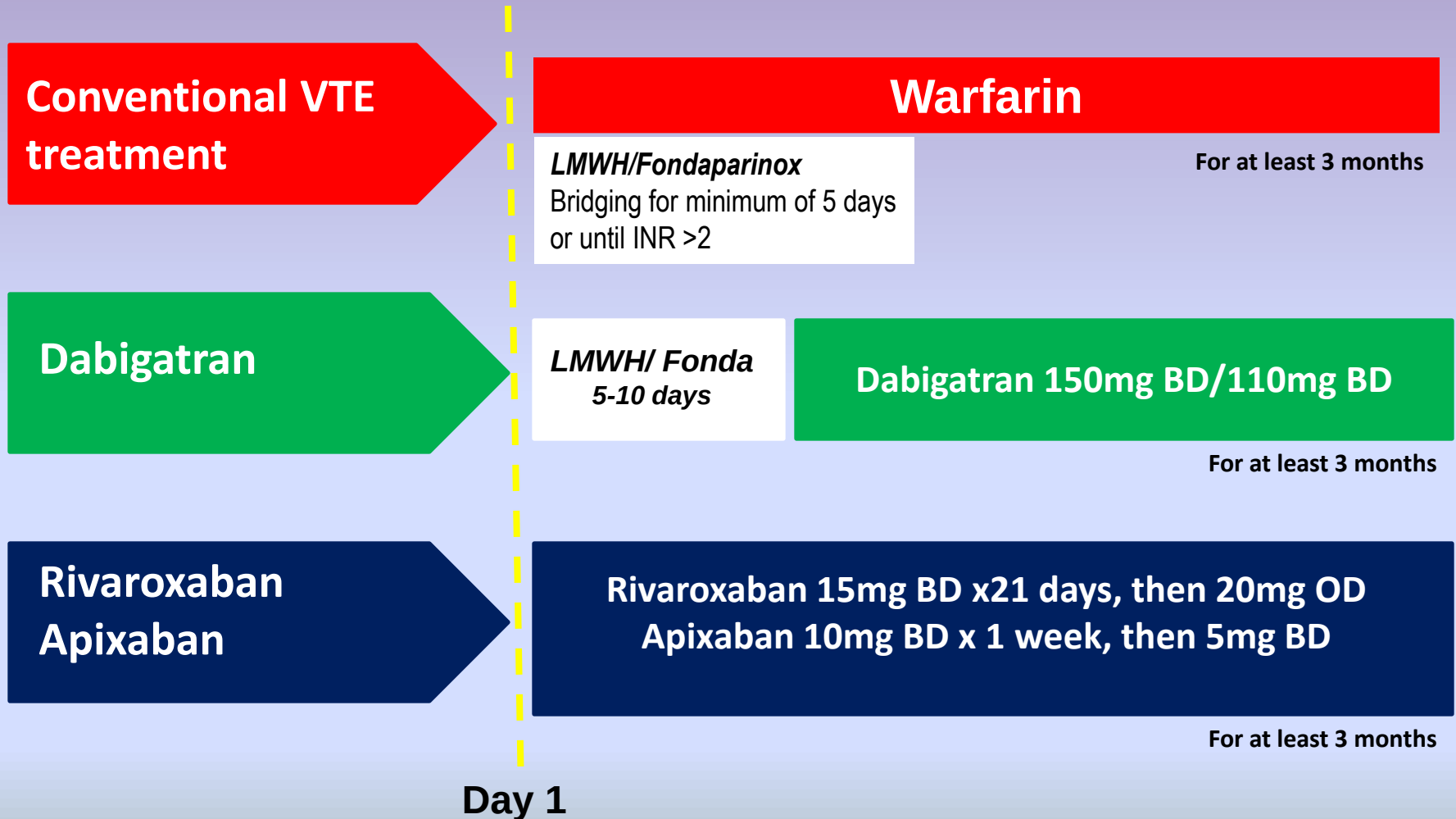
Indication	Dabigatran	Rivaroxaban	Apixaban
Atrial Fibrillation (lifelong)	•150mg BD •110mg BD*	•20mg OD •15mg OD¥	•5mg BD •2.5mg BD^δ
Venous Thromboembolism (minimum 3 months)	Parenteral treatment for 5-10 days , then 150mg BD (or 110mg BD*)	15mg BD for 21 days, then 20mg OD	10mg BD for 7 days, then 5mg BD

* For increased bleeding risk, age >80 years old, moderate renal impairment (CrCl 30-50ml/min), patients with gastritis or oesophagitis.

¥ For CrCl ≤ 50ml/min

^δ If patient has any 2 out of 3 of the following criteria: age ≥80 years, weight ≤60kg, Serum creat >133umol/L

Initiation of DOAC in acute VTE



Renal dosing in Atrial Fibrillation

CrCl	Dabigatran	Rivaroxaban	Apixaban
≥ 30ml/min	150mg BD	20mg OD (>50ml/min)	5mg BD
15-29ml/min	Not recommended	15mg OD (15-50ml/min)	2.5mg BD
<15ml/min		Not recommended	Not recommended

Reference:

-Product Leaflet for Dabigatran, Rivaroxaban, Apixaban

-Lexicomp

FDA: ESRF with HD 5mg BD,
2.5mg BD if age >80 yo;
weight ≤60kg

Renal dosing in VTE treatment

CrCl	Dabigatran	Rivaroxaban	Apixaban
≥30ml/min	No dosage adjustment	No dosage adjustment	No dosage adjustment
15-29ml/min	Avoid use	Use with caution	Use with caution
<15ml/min		Avoid use	Avoid use

Reference:

-Product insert for Dabigatran, Rivaroxaban, Apixaban

-Lexicomp

Hepatic Impairment

DOAC	Child Pugh Class	FDA Recommendation
Dabigatran	A	No dose adjustment
	B	Use with caution, no dose adjustment
	C	Not recommended
Rivaroxaban	A	No dose adjustment
	B	Not recommended
	C	Not recommended
Apixaban	A	No dose adjustment
	B	Use with caution, no dose adjustment
	C	Not recommended

Dabigatran and Apixaban : Patients with elevated liver enzymes ALT/AST>2x ULN were excluded in clinical trials.

Anticoagulant Conversion Guide

- [Reference\conversion\PRIC Anticoagulant conversion chart.xls](#)

Extreme Body Weight



What are the concerns when using DOAC in extreme body weight patient?

- Limited efficacy and safety data
- Inadequate drug exposure in obese patients
- Increased drug exposure in underweight patients

Dabigatran	< 50kg: Limited data. Risk of bleeding > 100kg: Concentration 20% lower
Rivaroxaban	<50kg. Cmax 24% higher
Apixaban	<50kg: 30% higher exposure. Risk of bleeding. >120kg: 30% lower exposure

Reference:

-Product insert for Dabigatran, Rivaroxaban, Apixaban

ISTH SSC - Guidance statements

- Recommend appropriate **standard dosing of the DOACs** in patients with a BMI less than or equal to 40 kg m^{-2} and weight less than or equal to 120 kg
- Suggest that DOACs **should not be used** in patients with a BMI of $> 40 \text{ kg m}^{-2}$ or a weight of $> 120 \text{ kg}$
- If DOACs are used in a patient with a BMI of $> 40 \text{ kg m}^{-2}$ or a weight of $> 120 \text{ kg}$, Suggest checking a **drug-specific peak and trough level**



Contents lists available at ScienceDirect

Pharmacological Research

journal homepage: www.elsevier.com/locate/yphrs



Direct oral anticoagulants in the treatment of acute venous thromboembolism in patients with obesity: A systematic review with meta-analysis

V. Mai^{a,b}, E. Marceau-Ferron^a, L. Bertoletti^{c,d,e,f}, Y. Lacasse^{a,g}, S. Bonnet^{a,b,g}, J.C. Lega^{f,h,i,l}, S. Provencher^{a,b,g,l,*}

Objectives: To compare the efficacy/safety of DOAC versus warfarin/LMWH for the treatment of VTE in patients with obesity and morbid obesity. The primary efficacy/safety outcomes were VTE recurrence and major bleeding.



Direct oral anticoagulants in the treatment of acute venous thromboembolism in patients with obesity: A systematic review with meta-analysis

V. Mai^{a,b}, E. Marceau-Ferron^a, L. Bertoletti^{c,d,e,f}, Y. Lacasse^{a,g}, S. Bonnet^{a,b,g}, J.C. Lega^{f,h,i,l}, S. Provencher^{a,b,g,l,*}

Results: We included 21 studies (50,360pts) of which 16150 patients had a BMI $\geq$ 30 kg/m² and 6443 patients had a BMI $\geq$ 40 kg/m². VTE recurrence was similar with DOAC compared to VKA/LMWH in patients with obesity (RR 1.03;95 %CI 0.93–1.15;p = 0.55) and morbid obesity (RR 1.06;95 %CI 0.94–1.19;p = 0.35). DOAC were also associated with a reduction in MB. Subgroup analyses comparing randomized controlled trials to observational studies showed consistent results. No difference was observed in regards of clinically relevant non-MB and mortality.

Conclusion: There is **no signal for a difference in VTE recurrence in patients with obesity and morbid obesity** treated by DOACs as compared to VKA/ LMWH. Therefore, DOAC appear to be as effective in preventing VTE recurrence while potentially reducing the risk of MB compared to VKA/ LMWH, supporting their use as first-line therapy for VTE. These results should however be **interpreted with caution given the overall low strength of evidence.**

In summary...

- If DOACs are to be used in patients with a BMI >40 or weight >120 kg, patients should be informed of the limitations of the available information and potential risk of under-dosing.
- There is no data on the best strategy for performing dose adjustments in obese patients.

Drug Interactions



Drug interactions

- All DOACs interact with **P-glycoprotein**.
- **CYP3A4** is an important metabolizer for apixaban (20-25%) and rivaroxaban (50%).
- In general, NOAC use is not recommended in combination with drugs that are strong inhibitors/ inducers of both CYP3A4 and P-gp. Such combinations should be avoided or used with great caution and surveillance.

Example of P-gp and CYP3A4 inhibitors and inducers:

	P-gp	CYP3A4
Inhibitors	Amiodarone, diltiazem, dronedarone, verapamil, macrolides, azoles	Amiodarone, clarithromycin, diltiazem, verapamil, azoles, protease inhibitors
Inducers	Rifampin, carbamazepine, phenytoin	Phenytoin, primidone, carbamazepine

DRUG INTERACTIONS THAT MAY AFFECT DOAC DRUG LEVELS

Potential ↑ in Apixaban		Potential ↓ in Apixaban	
<i>Diltiazem</i> *	<i>Naproxen</i> *	<i>Carbamazepine</i> ¥	
<i>Ketoconazole</i> ,	<i>Ritonavir</i> (all HIV	<i>Phenobarbital</i> ¥	
<i>itraconazole</i> ,	protease inhibitors)‡	<i>Phenytoin</i> ¥	
<i>voriconazole</i> ,	Strong inhibitors	<i>Rifampin</i> ¥	
<i>posaconazole</i> =	of both	<i>St. John's Wort</i> ¥	
<i>azole-antimycotics</i> ‡	<i>P-glycoprotein</i> and	Strong inducers of	
	<i>CYP 3A4</i> ‡	both <i>P-glycoprotein</i>	
		and <i>CYP-3A4</i> ¥	
Potential ↑ in Dabigatran		Potential ↓ in Dabigatran	
<i>Amiodarone</i> *	<i>Quinidine</i> *§	<i>Antacids</i> §	Strong
<i>Clarithromycin</i> *	<i>Ritonavir</i> *	<i>Atorvastatin</i> **	<i>P-glycoprotein</i>
<i>Cyclosporine</i> *	<i>Saquinavir</i> *	<i>Carbamazepine</i> ¥	inducers‡
<i>Dronedarone</i> ¥	<i>Tacrolimus</i> *	<i>Proton Pump</i>	<i>Phenytoin</i> ¥
<i>Itraconazole</i> *	<i>Tipranavir</i> ¥	<i>Inhibitors</i> *	
<i>Ketoconazole</i> ‡	<i>Ticagrelor</i> ¥	<i>Rifampin</i> ¥	
<i>Nelfinavir</i> *	<i>Verapamil</i> *§	<i>St. John's Wort</i> ¥	
<i>Posaconazole</i> *	Strong	<i>Tenofovir</i> ¥	
	<i>P-glycoprotein</i>		
	<i>inhibitors</i> ‡		
Potential ↑ in Rivaroxaban		Potential ↓ in Rivaroxaban	
<i>Clarithromycin</i> *	<i>Posaconazole</i> ‡	<i>Carbamazepine</i> ¥	Strong inducers of
<i>Erythromycin</i> *	<i>Ritonavir</i> ‡	<i>Phenobarbital</i> ¥	both <i>P-glycoprotein</i>
<i>Fluconazole</i> *	Strong inhibitors of	<i>Phenytoin</i> ¥	and <i>CYP 3A4</i> ¥
<i>Ketoconazole</i> ‡	both <i>P-glycoprotein</i>	<i>Rifampin</i> ¥	
<i>Itraconazole</i> ‡	and <i>CYP 3A4</i> ‡	<i>St. John's Wort</i> ¥	

Drug interactions

Anticoagulants and NSAIDs/Aspirin

- Coadministration of enoxaparin, warfarin, aspirin, clopidogrel and chronic NSAID use may increase the risk of bleeding.
- Avoid concurrent use of DOAC with other anticoagulants unless benefit outweighs risk.

Monitoring Parameters

- Pre-initiation:
 - ❑ Renal Profile (serum creatinine)
 - ❑ Liver Function Test (ALT, AST)
 - ❑ Full Blood Count (Hb, platelet)
 - ❑ Coagulation profile (PT, APTT)
 - ❑ Body weight
- Routinely monitored 1 to 3 months after initiation and then every 6 to 12 months thereafter.
- Periodic monitoring of hemoglobin, renal function and liver function, should be done at least annually or more frequently, where clinically indicated.

Administration

<i>Dabigatran</i>	•Must NOT be crushed, chewed or open capsules. •Must be stored in original packaging as light, moisture can cause product breakdown (Dabigatran is hygroscopic).
<i>Rivaroxaban</i>	•15mg and 20mg -Taken with food (AUC increases 39%, Cmax increases 75% with food). •For NG feeding, may be crushed and suspend in 50ml water, ensure NG tube not distal to stomach or decreased absorption can occur. •Crushed tablets in water are stable for up to 4 hours.
<i>Apixaban</i>	•May be taken with/without food. •For NG, may be crushed and suspend in 60ml water, D5W. •Crushed tablets in water are stable for up to 4 hours.

Reference: Product Insert for Dabigatran, Rivaroxaban, Apixaban

Counseling

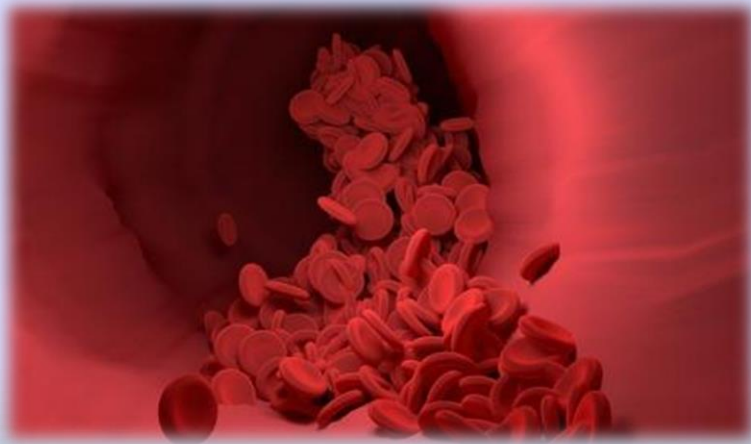
- ☐ Indication
- ☐ Duration of treatment
- ☐ How to take DOAC
- ☐ Missed dose
- ☐ Possible side effects
- ☐ Storage
- ☐ Plan for pregnancy
- ☐ Surgical/ procedure
- ☐ Sign & symptoms of Stroke/ DVT/ PE

Refer DOAC counseling checklist in Kaunseling folder



Symptoms of bleeding

- bruises
- Gum, nose bleed
- hemoptysis
- haematuria
- melaena ,red or black, tarry stool



Symptoms of renal failure

- excessive or rapid weight gain
- edema
- dehydration
- pruritus



Management of Bleeding and Reversal

Reversal of DOAC

- Cessation of DOACs may often be sufficient to reverse the anticoagulant effect in case of bleeding due to the short half life of DOAC.
- In severe or life-threatening hemorrhage, or emergency procedures on therapy, reversal of dabigatran anticoagulant effect with 5 grams of idarucizumab as a fixed-dose (intravenous infusion of two 2.5-gram) is recommended.
- Activated charcoal may also be used if the drug was ingested within last 2 to 4 hours (dabigatran), 1- 2 hours (rivaroxaban and apixaban).

- If idarucizumab is unavailable, then either PCC or activated prothrombin complex concentrate (aPCC) at 50U/kg (max dose 4,000 U) may be used.
- **Hemodialysis** has been suggested if dabigatran level is very high, especially in patients with impaired renal function.
- For reversal of rivaroxaban / apixaban, there are currently no specific antidotes clinically available. Coagulation factor supplementation with PCC or aPCC has been suggested as a potential nonspecific reversal strategy for the direct FXa inhibitors.

DOAC

Advantages	Limitations
☐ Fewer drug/food interaction ☐ Fewer laboratory monitoring and follow up ☐ Quick onset/offset of action (short half-lives) ☐ Relatively safe, proven to be at least non-inferior to warfarin and similar/improved bleed rates	☐ More expensive ☐ No specific reversal available (except for Dabigatran) ☐ Adherence--Missed dose can lead to under-anticoagulation, need full compliance

DOAC form in ehis

Direct Oral Anticoagulant Counseling Form

Encounter

Anticoagulation Plan

DOAC choice

Indication

Initiation date

Anticipated stop date

Dose

Planned duration

Status of anticoagulant :

Newly started ☐

Switch from oth anticoagulant ☐

Please specify

Continuation from current DOAC ☐

Baseline readings

Body weight kg

Laboratory

[Link Doc](#)

Creatinine Clearance (ml/min)

Co-morbid conditions

Concurrent Medications

Link Doc

Subjective Information

Compliance ----- Select ----- ▼

MyMAAT score

Bleeding ----- Select ----- ▼

Thrombosis ----- Select ----- ▼

Other complain/patients plan ----- Select ----- ▼

If yes, please specify

Counseling Checklist of DOAC

- ☐ : What is DOAC
- ☐ : Indication
- ☐ : Duration of treatment
- ☐ : How to take
- ☐ : Missed dose
- ☐ : Side effect
- ☐ : Monitoring of bleeding symptoms
- ☐ : Monitoring of renal failure symptoms

Pharmacist Review / Plan

Dose

Recommended dose

Drug/Dose

----- Select ----- ▼

----- Select -----

Maintain dose

Reduce dose

Coun Increase dose

** Advice patient to go to ED if had any bleeding / thrombosis symptoms.

Cost comparison

Dabigatran 110mg /150mg
(Pradaxa)

RM 152/30 tab

Rivaroxaban 15mg /20 mg
(Xarelto)

RM262/28 cap

Apixaban 2.5mg
(Eliquis)

RM174/30 cap

Apixaban 5mg
(Eliquis)

RM335/60 cap

Pharmaceutical Form

Dabigatran

Pradaxa hard cap 110 mg
[30's]



Pradaxa hard cap 75 mg
[30's]



Pradaxa hard cap 150 mg
[30's]



Rivaroxaban

Xarelto FC tab 15 mg
[28's]



Xarelto FC tab 20 mg
[28's]



Xarelto FC tab 10 mg
[10's]



Apixaban

Eliquis FC tab 2.5 mg
[30's]



Eliquis FC tab 5 mg
[20's]
[60's]



Take home message

**Always screen patient's
suitability before DOAC
counseling**

